

Hotel Booking Cancellation — Model Comparison Table

Binary Classification | Target: is_canceled | Dataset: Hotel Booking Demand (Kaggle)

Validation Set Results (sorted by ROC-AUC)

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.7907	0.7352	0.3761	0.4976	0.7879
Voting Ensemble	0.7877	0.7351	0.3591	0.4825	0.7867
Random Forest	0.7853	0.7402	0.3410	0.4669	0.7818
XGBoost	0.7770	0.6756	0.3676	0.4761	0.7684
SVC	0.7803	0.7745	0.2866	0.4184	0.7500
Bayesian (GNB)	0.7837	0.6773	0.4111	0.5117	0.7488
Decision Tree	0.7763	0.6893	0.3434	0.4584	0.7486
KNN	0.7580	0.6712	0.2394	0.3529	0.6932

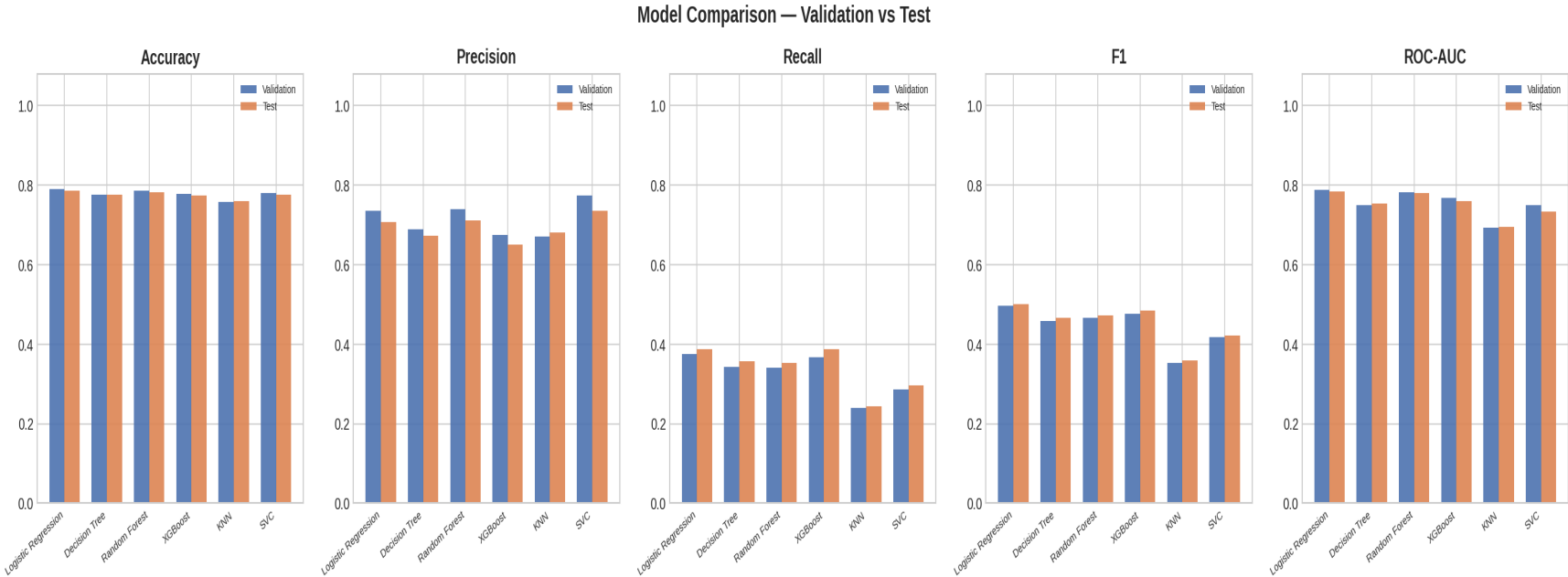
Test Set Results (sorted by ROC-AUC)

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.7867	0.7064	0.3869	0.5000	0.7832
Voting Ensemble	0.7823	0.6951	0.3748	0.4870	0.7821
Random Forest	0.7820	0.7105	0.3531	0.4717	0.7801
XGBoost	0.7737	0.6504	0.3869	0.4852	0.7602
Decision Tree	0.7750	0.6727	0.3579	0.4672	0.7529

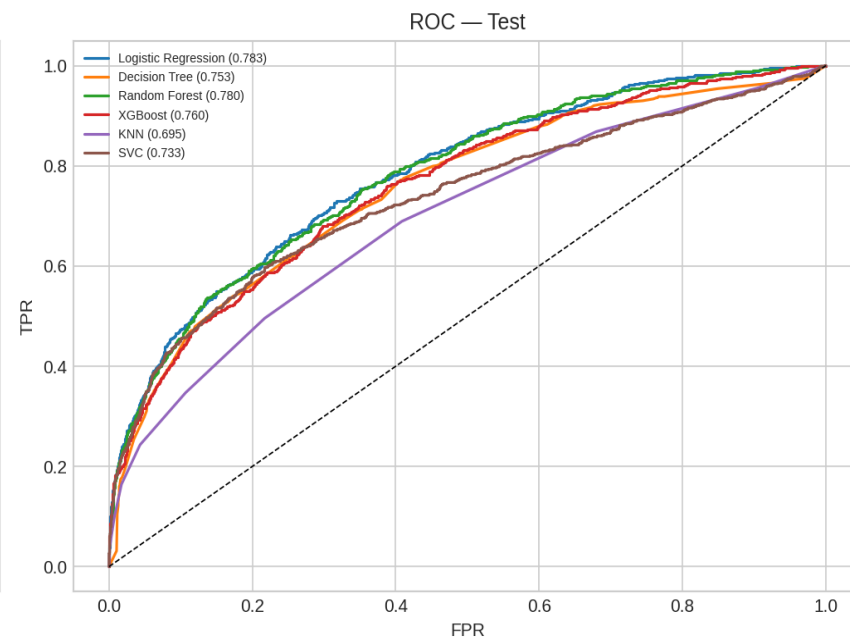
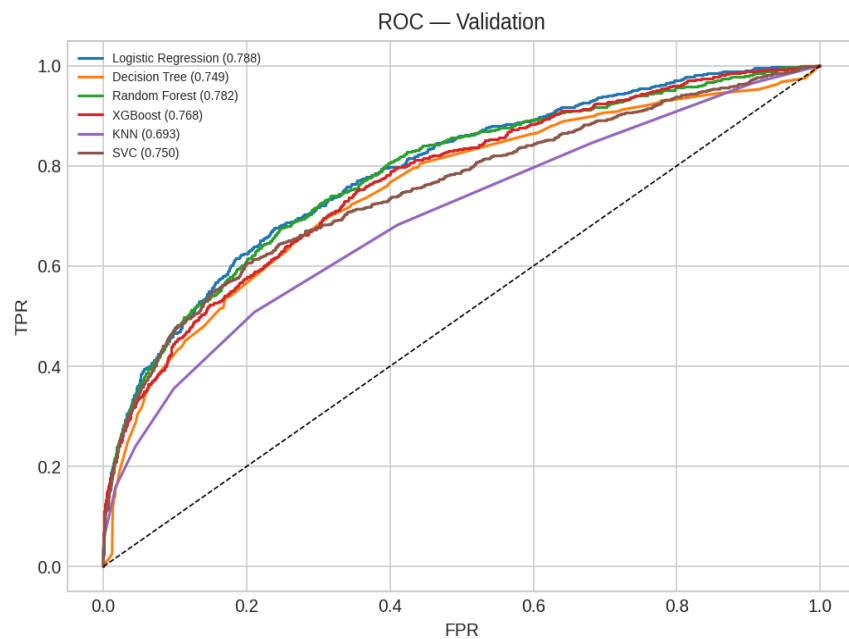
Model	Accuracy	Precision	Recall	F1	ROC-AUC
Bayesian (GNB)	0.7730	0.6327	0.4208	0.5054	0.7442
SVC	0.7767	0.7357	0.2963	0.4224	0.7331
KNN	0.7600	0.6814	0.2430	0.3583	0.6951

Blue highlighted row = best model by ROC-AUC. Ensemble rows at bottom of rankings.

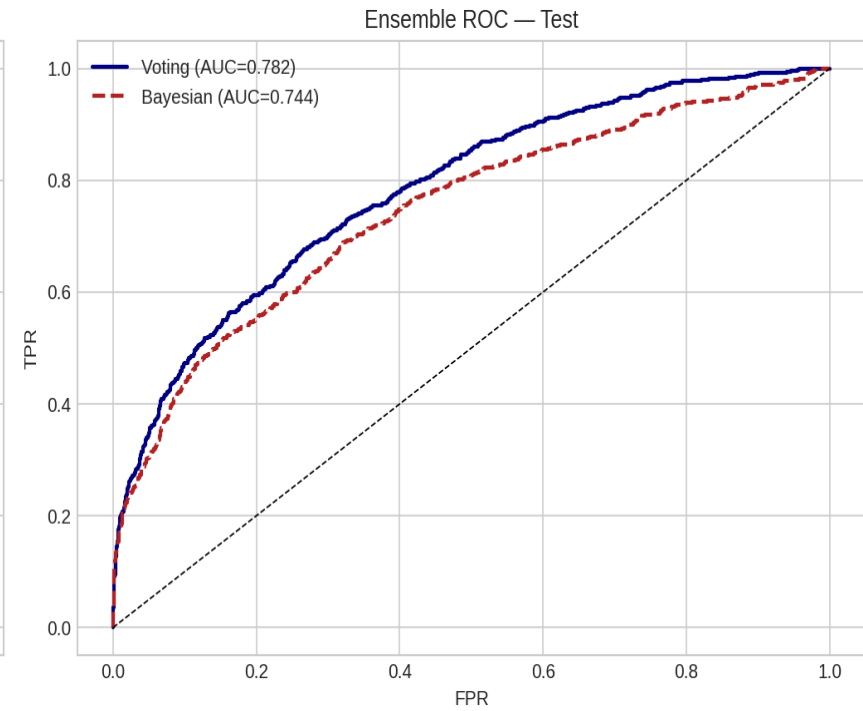
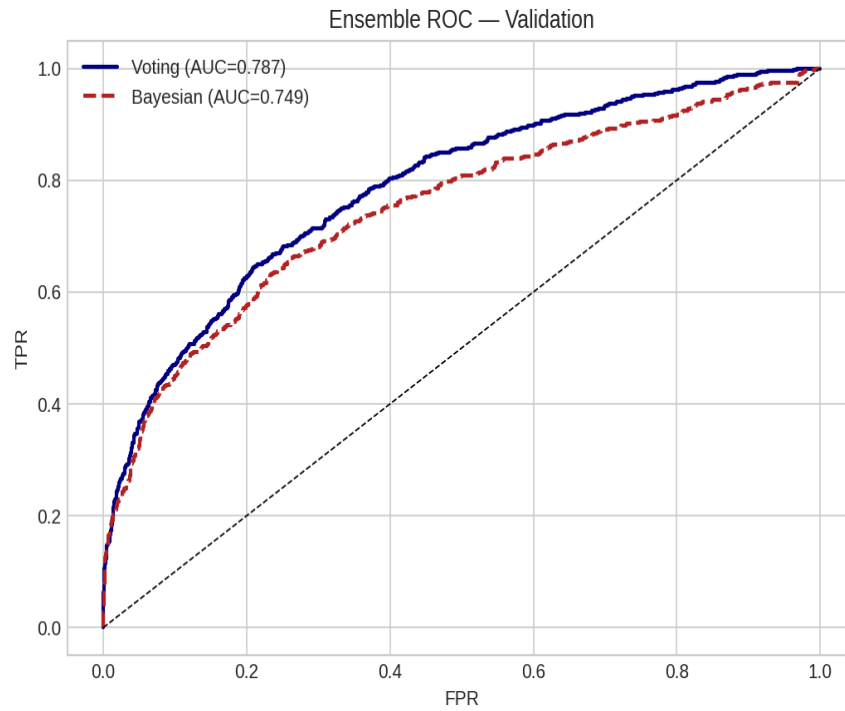
Visual Comparison — Validation vs Test



ROC Curves — All Models



Ensemble: Voting vs Bayesian ROC



Confusion Matrices — Top 3 Models (Validation)

Confusion Matrices — Validation (Top 3)

